KS3 DIGITAL TECHNOLOGY

OPERATING SYSTEMS

The software that runs everything else: what an operating system does, how it juggles processes and memory, how files and the command line work, why updates and permissions keep you safe, how open-source Linux powers the world, and how AI is reshaping the operating systems of the future.

MR BUTLER'S DIGITAL TECHNOLOGY

LESSONS

1 What does an OS do?

2 Processes and memory

3 Files and the command line

4 Updates, accounts and security

5 Open source and Linux

6 The future of operating systems

LESSON 1 OF 6

WHAT DOES AN OS DO?

An operating system (OS) is system software that manages a computer's hardware and software and provides a user interface.

STARTER · 5 MINUTES

DO NOW

What would happen if every app on your phone tried to use the screen, speaker and camera at the same time with no one in charge?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Define an operating system, describe its main functions and give examples for different devices

SUCCESS CRITERIA: I CAN...

- ✓ I can define an operating system
- ✓ I can describe the main functions of an operating system
- ✓ I can name operating systems for different devices

THE BOSS OF THE COMPUTER

- An operating system (OS) is the software that manages everything else
- Sits between the hardware and your apps
- Apps ask the OS for memory, storage, the screen and the internet
- Without an OS, every app would need to control the hardware itself

KEY WORDS

OPERATING SYSTEM	System software that manages a computer's hardware and software and provides a user interface.
SYSTEM SOFTWARE	Software that runs and manages the computer itself, rather than doing a task for the user.
USER INTERFACE	The way a person interacts with a computer, such as windows, icons, touch or voice.
DEVICE DRIVER	Software that lets the operating system control a piece of hardware.
BOOTING	Starting a computer and loading the operating system into memory.

JOBS OF AN OS

- Process management: shares out the CPU between programs
- Memory management: decides what's in RAM
- File management: organises storage into files and folders
- Device management (drivers), security and the user interface

ACTIVITY · 15 MINUTES · PAIRS

OS SPOTTER

1. Make a list of every device you and your family use that has an operating system (phones, laptops, consoles, TVs, cars, watches, routers).
2. For each, name the OS if you know it.
3. Choose one device and describe three jobs its OS does for you.

In your workbook, answer:

? Describe three jobs your phone's OS does.

? Why do you think so many different devices need an operating system?

Your workbook: Lesson 1 · tick each step off as you go

OSS EVERYWHERE

- Desktops and laptops: Windows, macOS, Linux, ChromeOS
- Phones and tablets: Android, iOS, iPadOS
- Consoles, smart TVs, cars and watches all have their own
- Tiny embedded systems may run a real-time OS

QUICK CHECK

1. What is an operating system?

A A website
B Software that manages a computer's hardware and software
C A type of CPU
D A word processor

2. What is a device driver?

A A screwdriver
B A type of RAM
C A person who delivers devices
D Software that lets the OS control a piece of hardware

3. Which is a phone operating system?

A Chrome browser
B Minecraft
C Android
D Microsoft Word

4. What is booting?

A Kicking a computer
B Starting a computer and loading the OS
C Deleting files
D Shutting down

QUICK CHECK: ANSWERS

1. **B: Software that manages a computer's hardware and software**
   The OS is in charge.

2. **D: Software that lets the OS control a piece of hardware**
   Drivers talk to hardware.

3. **C: Android**
   Also iOS.

4. **B: Starting a computer and loading the OS**
   Loads the OS into memory.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Define the term operating system.

Look back at today's success criteria: can you tick them all?

- ✓ I can define an operating system
- ✓ I can describe the main functions of an operating system
- ✓ I can name operating systems for different devices

LESSON 2 OF 6

PROCESSES AND MEMORY

A running program is a process.

DO NOW

Open Task Manager (Ctrl + Shift + Esc). How many processes are running? Did you start them all?

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how an operating system manages processes, multitasking and memory

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what a process is and how multitasking works
- ✓ I can explain how the OS manages RAM and virtual memory
- ✓ I can use Task Manager to investigate processes

PROCESSES

- A process is a program that is running
- One app can have many processes (each browser tab may be one)
- Background processes: updates, antivirus, syncing
- Task Manager (Windows) and Activity Monitor (Mac) show them

KEY WORDS

PROCESS	A program that is currently running.
MULTITASKING	Running more than one program at the same time.
SCHEDULING	How the operating system decides which process uses the CPU and for how long.
RAM	Random access memory: fast, temporary memory that holds programs and data in use.
VIRTUAL MEMORY	Part of storage used as extra memory when RAM is full.

MULTITASKING

- A CPU core runs one thing at a time
- The OS scheduler gives each process a tiny time slice, thousands of times a second
- It looks like everything is running at once
- Multi-core CPUs really can run several at once

MEMORY MANAGEMENT

- RAM holds running programs and their data
- The OS gives each process its own space, so they can't interfere
- When RAM is full, virtual memory uses storage as overflow
- Storage is much slower than RAM, so the computer slows down

ACTIVITY · 20 MINUTES · PAIRS

TASK MANAGER INVESTIGATION

1. Open Task Manager (Ctrl + Shift + Esc) and choose "More details" if needed.
2. Record the CPU and memory use. Then open five browser tabs, a video and a document, and record again.
3. Find the three processes using the most memory.
4. Explain what you found using the words process, scheduling and RAM.

In your workbook, answer:

? How did CPU and memory use change, and why?

? Why might a computer slow down when lots of apps are open?

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. What is a process?

A A cable
B A type of storage
C A program that is currently running
D A file type

2. How does a single CPU core multitask?

A It uses the printer
B It runs everything at exactly the same time
C It doesn't
D The OS switches between processes very quickly

3. What is virtual memory?

A Storage used as overflow when RAM is full
B Cloud storage
C Memory in VR headsets
D Extra RAM chips

4. Why does heavy use of virtual memory slow a computer down?

A Storage is much slower than RAM
B It turns off the CPU
C Storage is faster
D It deletes files

QUICK CHECK: ANSWERS

1. C: A program that is currently running

Running programs are processes.

2. D: The OS switches between processes very quickly

Time slicing.

3. A: Storage used as overflow when RAM is full

It's slower than RAM.

4. A: Storage is much slower than RAM

Slower overflow.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how an operating system allows several programs to appear to run at once on one CPU core.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what a process is and how multitasking works
- ✓ I can explain how the OS manages RAM and virtual memory
- ✓ I can use Task Manager to investigate processes

LESSON 3 OF 6

FILES AND THE COMMAND LINE

The OS's file system organises storage into a hierarchy of folders (directories) and files, each with a path.

STARTER · 5 MINUTES

DO NOW

Where is the last file you saved? Can you describe exactly how to get to it?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how file systems organise data and use basic command-line instructions

SUCCESS CRITERIA: I CAN...

- ✓ I can describe how folders and files are organised in a hierarchy
- ✓ I can write a file path and explain file extensions
- ✓ I can use basic command-line instructions

FILES AND FOLDERS

- The file system organises storage into folders (directories) and files
- Folders inside folders make a hierarchy, like an upside-down tree
- The top is the root (e.g. C:\ on Windows, / on Mac and Linux)
- Each file has a path showing exactly where it is

KEY WORDS

FILE SYSTEM	The part of the operating system that organises data on storage into files and folders.
DIRECTORY	Another name for a folder.
FILE PATH	The full location of a file, listing each folder from the top level down.
FILE EXTENSION	The letters after the dot in a file name, showing the type of file.
COMMAND-LINE INTERFACE	A way of controlling a computer by typing text commands.

PATHS AND EXTENSIONS

- Windows: C:\Users\Sam\Documents\Homework\essay.docx
- Mac/Linux: /home/sam/Documents/Homework/essay.docx
- The extension shows the file type: .docx, .pdf, .jpg, .mp4, .py, .exe
- Be careful: .exe files run programs and can be malware

THE COMMAND LINE

- A command-line interface (CLI): type commands instead of clicking
- Faster for some jobs; can automate repetitive tasks
- Used by programmers, IT technicians and on servers with no screen
- Windows: Command Prompt/PowerShell; Mac/Linux: Terminal

ACTIVITY · 25 MINUTES · PAIRS

COMMAND-LINE EXPLORER

1. Open Command Prompt (search "cmd") or Terminal.

2. Use dir/ls, cd and mkdir to create a folder called "OSlesson" in your Documents and a subfolder called "test".

3. Write down the full file path of your "test" folder.

4. Complete the treasure hunt: follow your teacher's path instructions to find the hidden file.

In your workbook, answer:

? Write the commands you used to create OSlesson\test.

? Give one reason a technician might prefer the command line to a graphical interface.

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. What is a directory?

A A file type
B A CPU part
C Another name for a folder
D A phone book only

2. What does a file extension show?

A The date
B The type of file
C Who made it
D The file's size

3. Which command lists files in Windows Command Prompt?

A ls only
B show
C dir
D list

4. Why is "homework.pdf.exe" dangerous?

A It's a large PDF
B It's encrypted
C It's a video
D It's a program disguised as a PDF

QUICK CHECK: ANSWERS

1. **C: Another name for a folder**
   Folders are directories.

2. **B: The type of file**
   e.g. .docx, .jpg.

3. **C: dir**
   ls is used on Mac/Linux.

4. **D: It's a program disguised as a PDF**
   Double extensions hide malware.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Write the file path for a file called notes.txt in a folder called Science inside Documents for user Ali on Windows.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe how folders and files are organised in a hierarchy
- ✓ I can write a file path and explain file extensions
- ✓ I can use basic command-line instructions

LESSON 4 OF 6

UPDATES, ACCOUNTS AND SECURITY

OS updates fix bugs and security holes, so installing them quickly is vital; when an OS reaches end of support it stops getting fixes.

STARTER · 5 MINUTES

DO NOW

Your laptop says "Updates ready: restart now?" Do you click "Later"? Why might that be risky?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how operating system updates, user accounts, permissions and built-in security features protect a computer

SUCCESS CRITERIA: I CAN...

- ✓ I can explain why OS updates matter, including end of support
- ✓ I can explain user accounts, administrator rights and permissions
- ✓ I can describe built-in OS security features

UPDATES AND PATCHES

- Updates add features and fix bugs
- Security patches fix vulnerabilities attackers could use
- Once a fix is public, attackers race to use the hole on unpatched computers
- Turn on automatic updates and restart when asked

KEY WORDS

PATCH	An update that fixes a bug or security vulnerability.
VULNERABILITY	A weakness in software that an attacker could exploit.
END OF SUPPORT	When a software maker stops releasing updates and security fixes.
ADMINISTRATOR	A user account that can install software and change system settings.
PERMISSIONS	Rules that control what a user or app is allowed to do.

END OF SUPPORT

- Every OS version eventually stops getting updates
- Windows 10 reached end of support in October 2025
- Unsupported systems become more dangerous over time
- Options: upgrade, pay for extended updates, or switch OS

ACCOUNTS AND PERMISSIONS

- Standard users can't change important system settings
- Administrators can install software and change settings
- Use a standard account day to day; admin only when needed
- App permissions: camera, microphone, location, contacts

ACTIVITY · 20 MINUTES · PAIRS

PERMISSION PATROL

1. Look at the list of 8 (fictional) apps and the permissions they request (e.g. a torch app wanting your contacts and location).
2. For each, decide which permissions are reasonable and which are suspicious.
3. Look at the update history of a school computer (screenshot) and identify any risks.
4. Write a 5-point security checklist for a new laptop.

In your workbook, answer:

? Which app requests are suspicious, and why?

? Write your 5-point checklist for a new laptop.

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What is a patch?

A An update that fixes a bug or vulnerability
B A cable
C A sticker
D A type of virus

2. What does end of support mean?

A The computer breaks
B Support staff go home
C The maker stops releasing security updates
D The OS becomes free

3. Why use a standard account rather than administrator for everyday use?

A Malware can't easily change the whole system
B It saves battery
C It's faster
D Admin accounts can't browse

4. What is sandboxing?

A Running apps in a restricted area so they can't access other apps' data
B Playing in sand
C Backing up files
D Deleting apps

QUICK CHECK: ANSWERS

1. **A: An update that fixes a bug or vulnerability**
Patches fix holes.

2. **C: The maker stops releasing security updates**
No more fixes.

3. **A: Malware can't easily change the whole system**
Least privilege.

4. **A: Running apps in a restricted area so they can't access other apps' data**
Contains damage.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why it's risky to delay installing security updates.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain why OS updates matter, including end of support
- ✓ I can explain user accounts, administrator rights and permissions
- ✓ I can describe built-in OS security features

LESSON 5 OF 6

OPEN SOURCE AND LINUX

Proprietary software is owned by a company that keeps its code secret, while open-source software's code is public and can be studied, changed and shared.

STARTER · 5 MINUTES

DO NOW

Would you trust a recipe more if you could read every ingredient, or one that's kept secret? Does the same apply to software?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Compare open-source and proprietary software and describe the role of Linux

SUCCESS CRITERIA: I CAN...

- ✓ I can explain the difference between open-source and proprietary software
- ✓ I can describe where Linux is used and why
- ✓ I can evaluate which type of software suits a situation

OPEN VS PROPRIETARY

- Source code: the human-readable instructions programmers write
- Proprietary: owned by a company; code kept secret (Windows, macOS, iOS)
- Open source: code is public; anyone can study, change and share it
- Open source isn't always free, and free isn't always open source

KEY WORDS

OPEN SOURCE	Software whose source code is public and can be studied, changed and shared.
PROPRIETARY SOFTWARE	Software owned by a company that keeps its source code secret.
SOURCE CODE	The human-readable instructions programmers write.
LINUX	A free, open-source operating system first created by Linus Torvalds in 1991.
LICENCE	A legal agreement about how software can be used, copied and changed.

LINUX RUNS THE WORLD

- Created by Linus Torvalds in 1991 as a student project
- Runs most web servers and cloud computers
- Runs all of the world's top 500 supercomputers
- Android is built on the Linux kernel; so are many routers, TVs and cars

PROS AND CONS

- Open source: free, flexible, transparent, huge community
- But support can be harder to find, and some software doesn't run on it
- Proprietary: polished, official support, wide compatibility
- But costs money and you can't see or change the code

ACTIVITY · 25 MINUTES · GROUPS OF 3

CHOOSE THE SOFTWARE

1. Read the four scenarios: a charity reviving 50 old laptops; a hospital needing official support; a games streamer; a start-up building a website on cloud servers.

2. For each, decide whether to use open-source or proprietary software (or both) and choose an OS.

3. Justify each choice with at least two reasons.

4. Try a Linux distro if available (on a Raspberry Pi or in a browser-based Linux demo) and compare it with Windows.

In your workbook, answer:

? What would you choose for the charity, and why?

? What would you choose for the hospital, and why?

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. What is open-source software?

A Software with secret code
B Software that's always paid
C A type of virus
D Software whose source code is public and can be changed and shared

2. Which is proprietary software?

A Linux
B LibreOffice
C Windows
D Firefox

3. Who created Linux?

A Steve Jobs
B Bill Gates
C Alan Turing
D Linus Torvalds

4. Which phone OS is built on the Linux kernel?

A Windows Phone
B iOS
C None
D Android

QUICK CHECK: ANSWERS

1. **D: Software whose source code is public and can be changed and shared**

   Open code.

2. **C: Windows**

   Owned by Microsoft.

3. **D: Linus Torvalds**

   In 1991.

4. **D: Android**

   Android uses Linux.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain the difference between open-source and proprietary software.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain the difference between open-source and proprietary software
- ✓ I can describe where Linux is used and why
- ✓ I can evaluate which type of software suits a situation

LESSON 6 OF 6

THE FUTURE OF OPERATING SYSTEMS

Operating systems are being reshaped by AI: new computers and phones have NPUs (neural processing units) so AI features run on the device, and OSs are adding AI assistants that can act across apps.

STARTER · 5 MINUTES

DO NOW

Imagine telling your laptop "Find the photo of my dog from last summer and send it to Gran" and it just does it. What would the OS need to be able to do?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how operating systems are changing with on-device AI, the cloud and new kinds of devices, and evaluate the impacts

SUCCESS CRITERIA: I CAN...

- ✓ I can describe how AI is being built into operating systems
- ✓ I can describe cloud-based and new-device operating systems
- ✓ I can evaluate the benefits and risks of AI-powered operating systems

AI INSIDE THE OS

- New chips include an NPU (neural processing unit) for AI tasks
- On-device AI: features run locally, not in the cloud
- Search your files by describing them; live captions and translation
- AI assistants built into Windows, macOS, iOS and Android

KEY WORDS

NPU	Neural processing unit: a chip designed to run AI tasks efficiently.
ON-DEVICE AI	AI that runs on the device itself rather than in the cloud.
AI AGENT	AI that can carry out multi-step tasks on a user's behalf.
CLOUD-BASED OS	An operating system that runs most apps and stores most files online.

FROM ASSISTANT TO AGENT

- AI agents can carry out multi-step tasks across apps for you
- "Book the cheapest train and add it to my calendar"
- The OS must manage what the AI is allowed to see and do
- Permissions and human approval become even more important

NEW KINDS OF OS

- Cloud-based OSs: ChromeOS runs mostly in the browser
- Headset OSs for mixed reality, e.g. visionOS and Android XR
- Car OSs that update over the air
- Tiny OSs for billions of IoT devices

ACTIVITY · 30 MINUTES · GROUPS OF 3

OS OF 2035

1. Design the operating system for a device in 2035 (glasses, a car, a home robot, or a school laptop).

2. Describe three AI features it would have and how they help users.

3. Write the permission rules: what the AI can do automatically, what needs approval, and what it can never do.

4. Present your OS and explain how it protects privacy and security.

In your workbook, answer:

? Describe your three AI features.

? Write your permission rules.

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What is an NPU?

A A type of RAM
B A chip designed to run AI tasks efficiently
C A screen
D A network cable

2. What is one benefit of on-device AI?

A It's slower
B Data can stay on the device, improving privacy
C It uses the cloud only
D It always needs the internet

3. What is an AI agent?

A A secret agent
B AI that can carry out multi-step tasks across apps for you
C A type of keyboard
D A virus

4. Which OS runs most apps in a web browser?

A ChromeOS
B iOS
C DOS
D Windows 95

QUICK CHECK: ANSWERS

1. **B: A chip designed to run AI tasks efficiently**
Neural processing unit.

2. **B: Data can stay on the device, improving privacy**
Local processing.

3. **B: AI that can carry out multi-step tasks across apps for you**
It acts on your behalf.

4. **A: ChromeOS**
Cloud-based OS.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain two advantages of on-device AI compared with cloud AI.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe how AI is being built into operating systems
- ✓ I can describe cloud-based and new-device operating systems
- ✓ I can evaluate the benefits and risks of AI-powered operating systems

END OF TOPIC

OPERATING SYSTEMS

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.